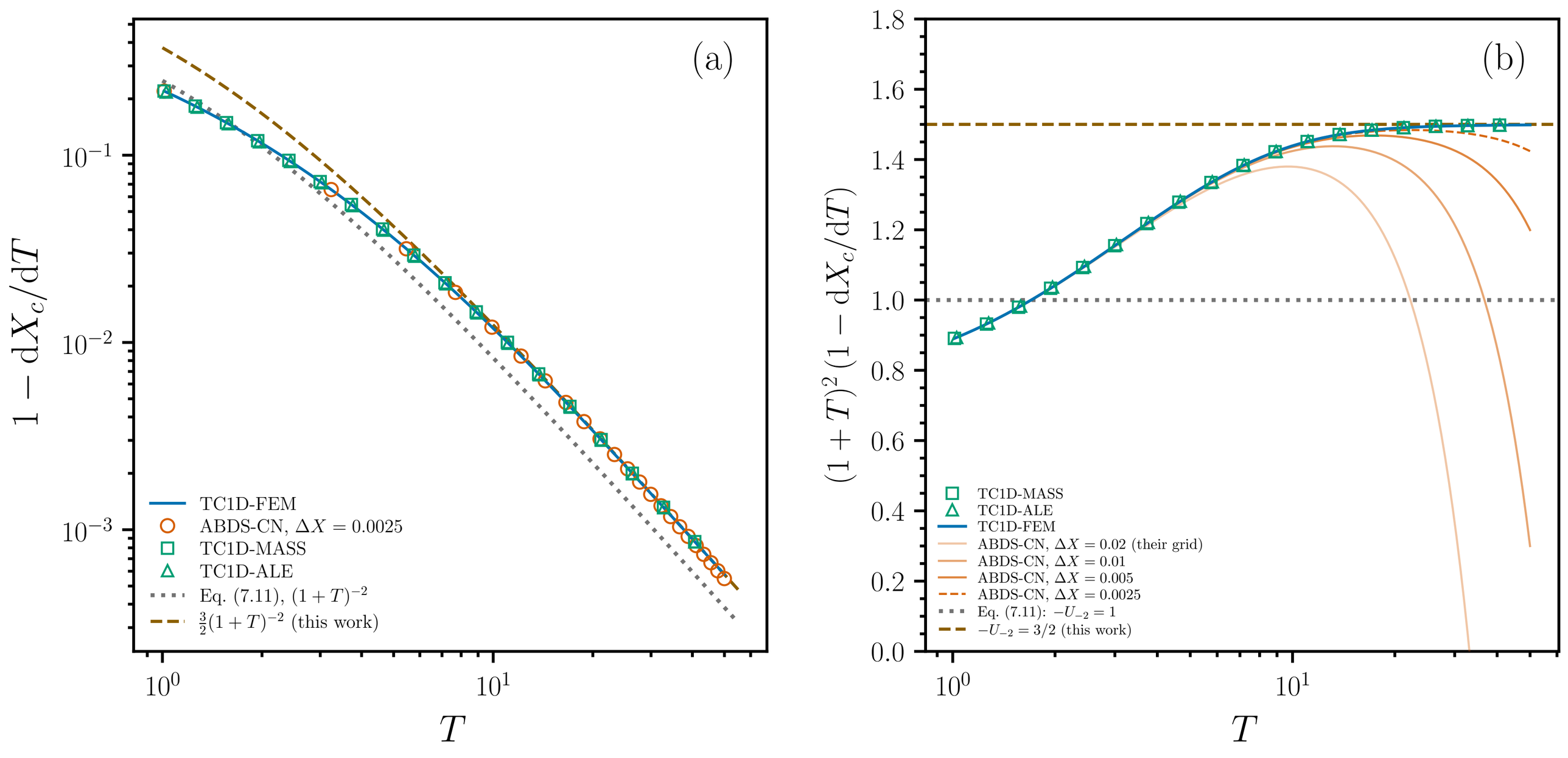

(a)
$1 - \mathrm{d}X_c/\mathrm{d}T$
10^{-1}
10^{-2}
10^{-3}
10^{0}
10^{1}
T
TC1D-FEM
ABDS-CN, $\Delta X = 0.0025$
TC1D-MASS
TC1D-ALE
Eq. (7.11), $(1+T)^{-2}$
$\frac{3}{2}(1+T)^{-2}$ (this work)
(b)
$(1+T)^2(1-\mathrm{d}X_c/\mathrm{d}T)$
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
10^{0}
10^{1}
T
TC1D-MASS
TC1D-ALE
TC1D-FEM
ABDS-CN, $\Delta X = 0.02$ (their grid)
ABDS-CN, $\Delta X = 0.01$
ABDS-CN, $\Delta X = 0.005$
ABDS-CN, $\Delta X = 0.0025$
Eq. (7.11): $-U_{-2} = 1$
$-U_{-2} = 3/2$ (this work)